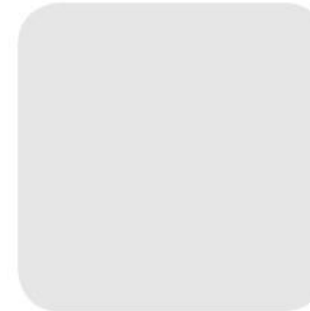
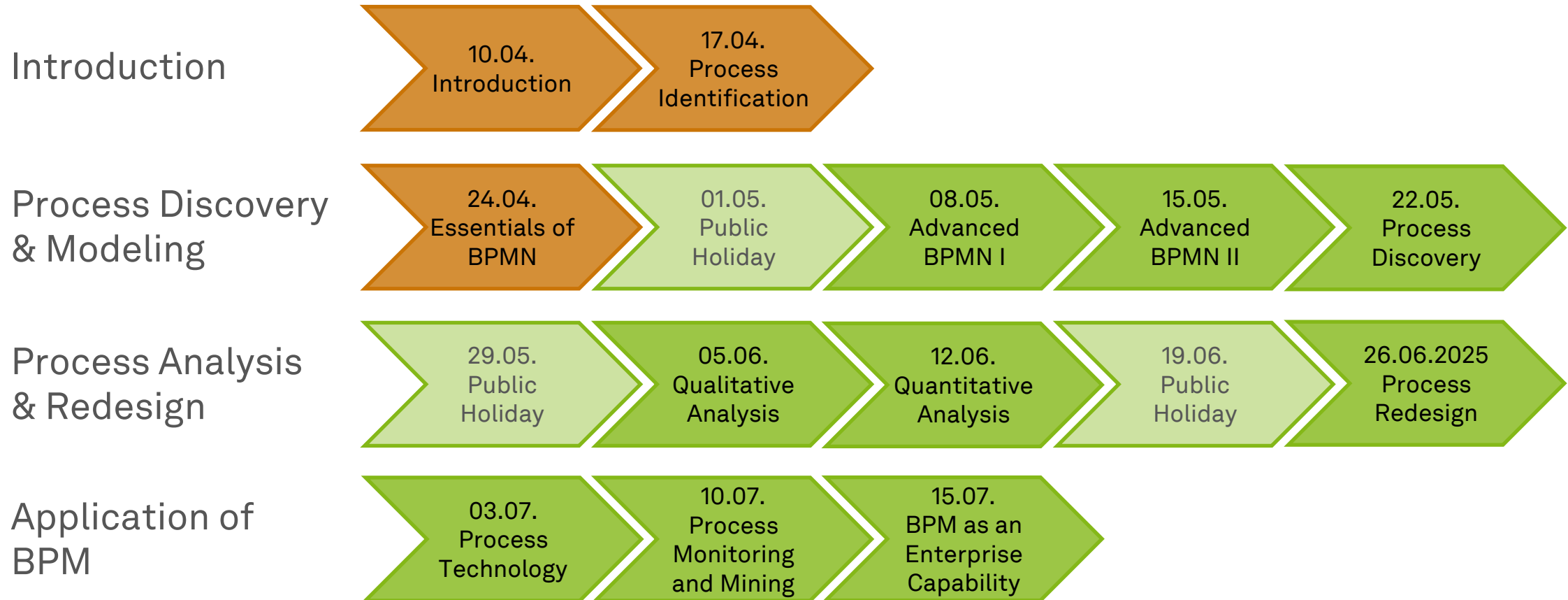


Business Process Management

Advanced Process Modeling I



Roadmap



Learning Goals of Today

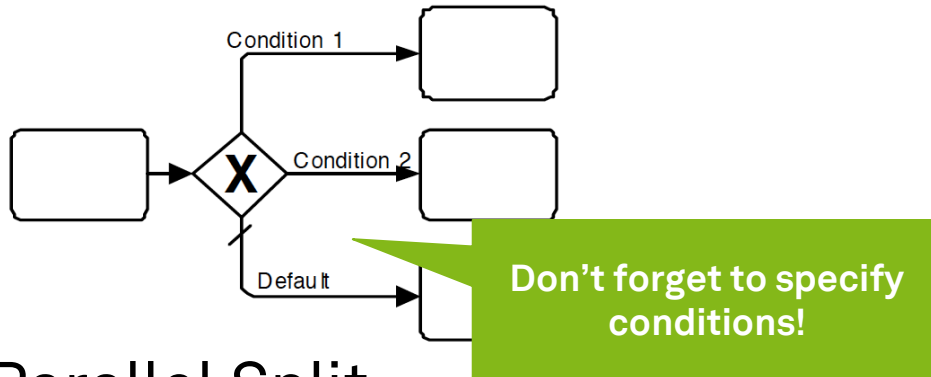
- Better understand the syntactics and semantics of BPMN
- Understand branching and merging in BPMN
- Understand decomposition, resources, and collaborations in BPMN
- Be able to use advanced BPMN

Branching & Merging

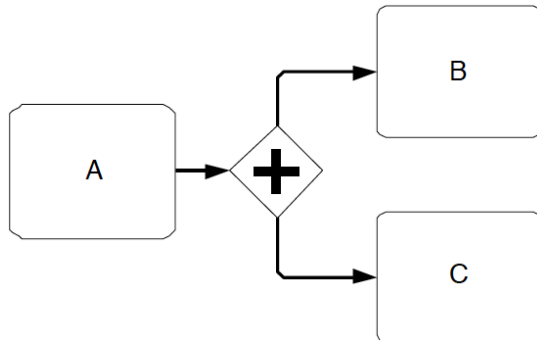


Gateways in Action

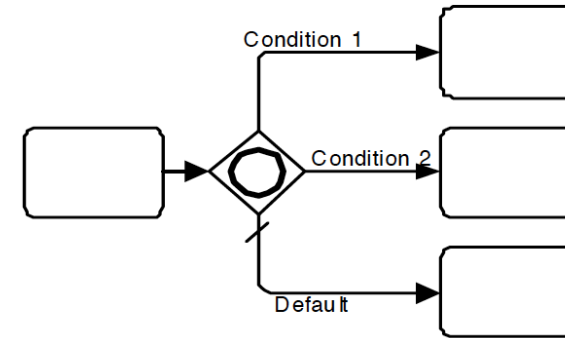
- Exclusive Decision



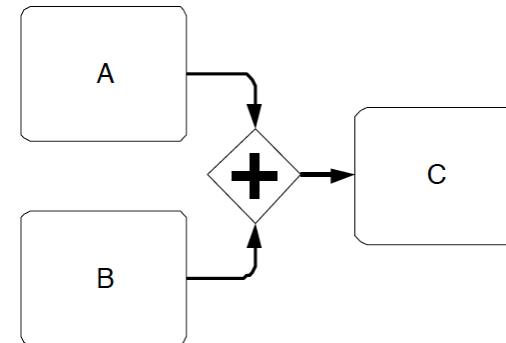
- Parallel Split



- Inclusive Decision

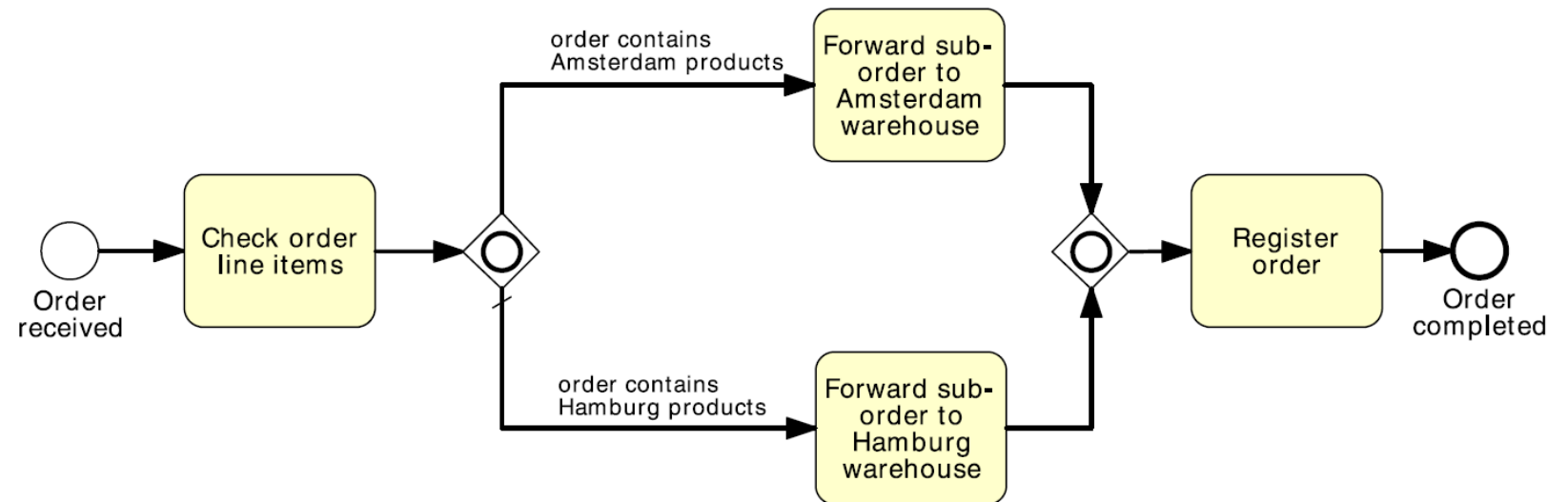


- Synchronizing Merge



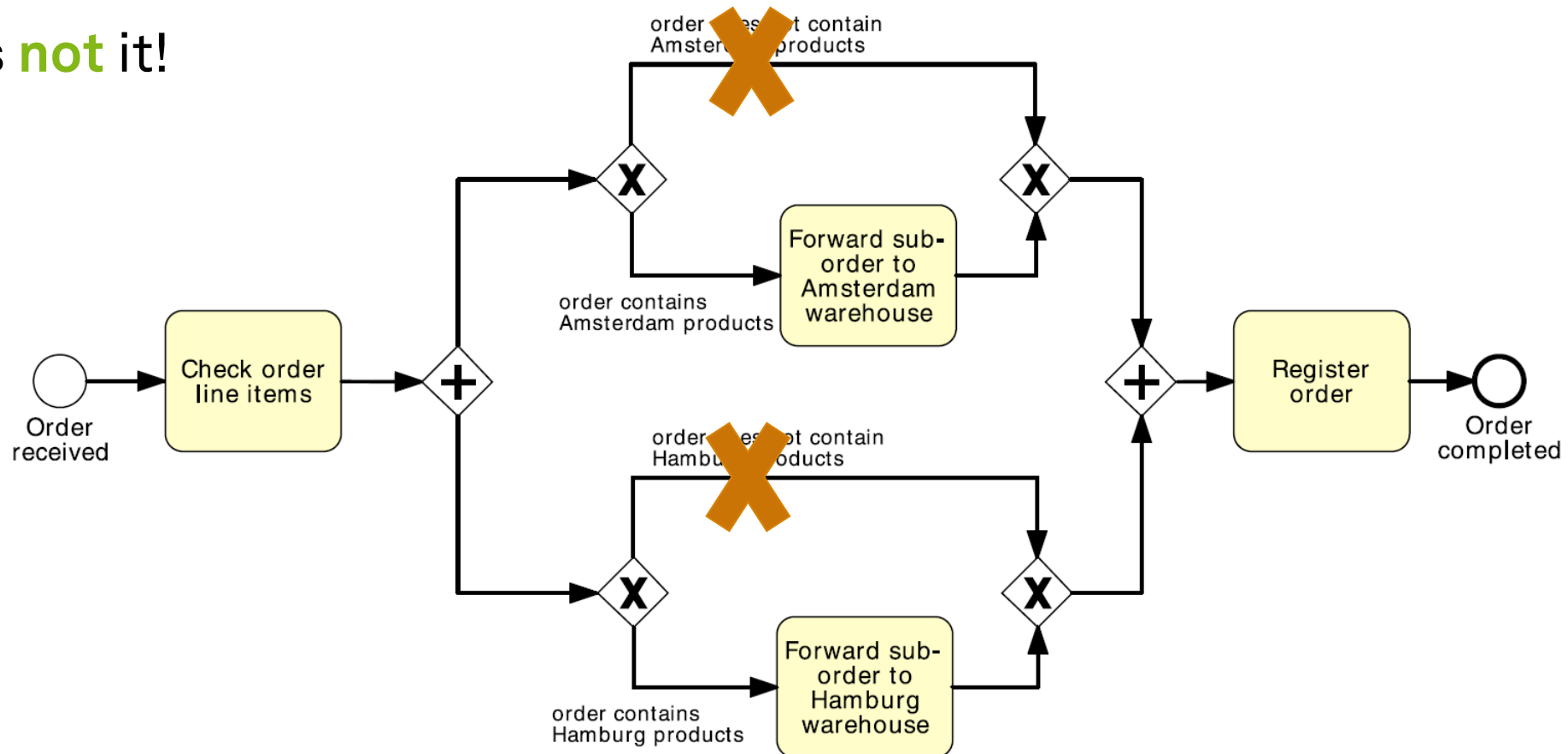
Do We Really Need an Inclusive OR?

- Imagine this process
- You can visit **one or both** branches
- Can you model it **without an OR gateway**?



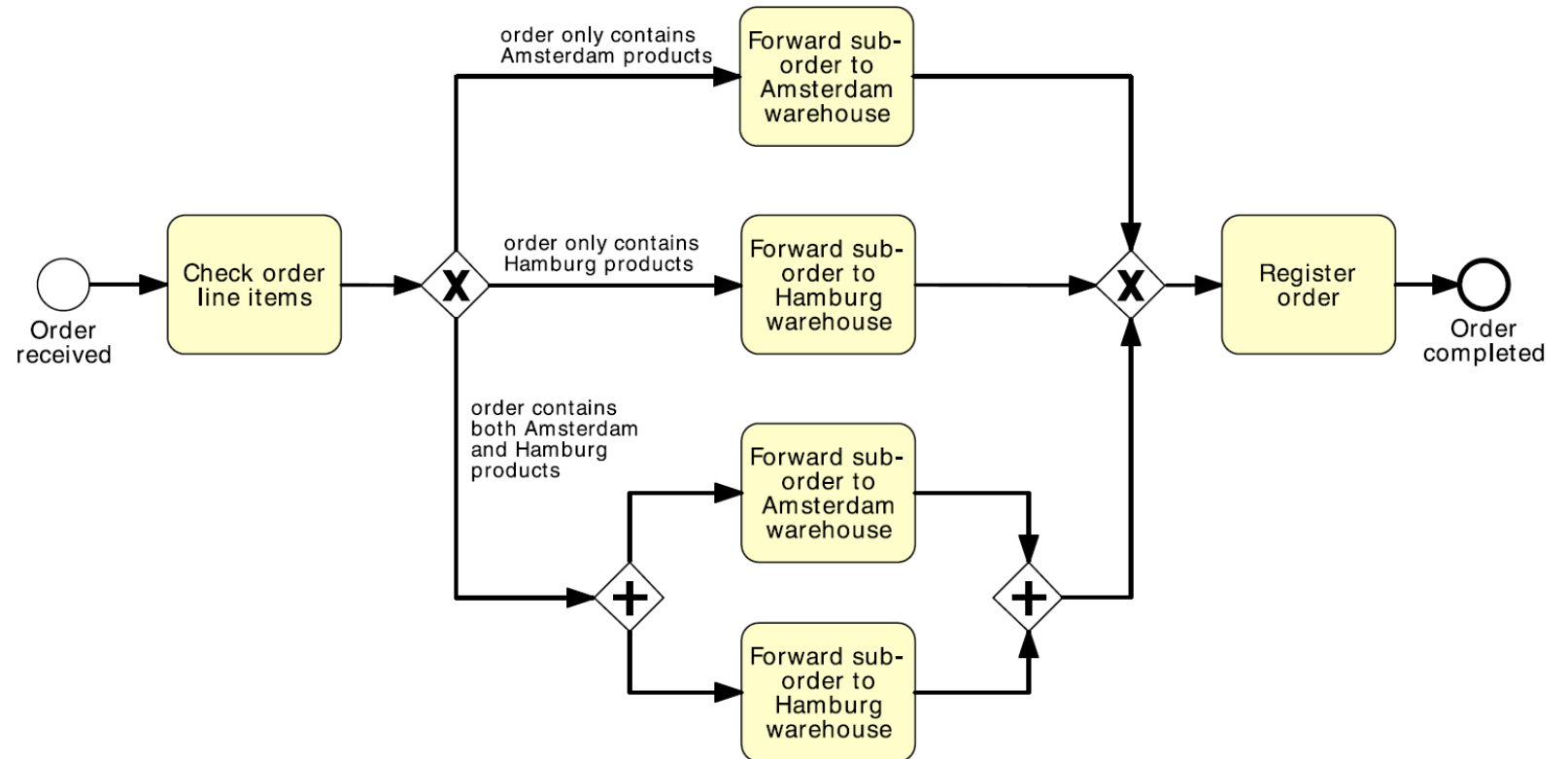
A Solution Without OR?

- This is **not** it!

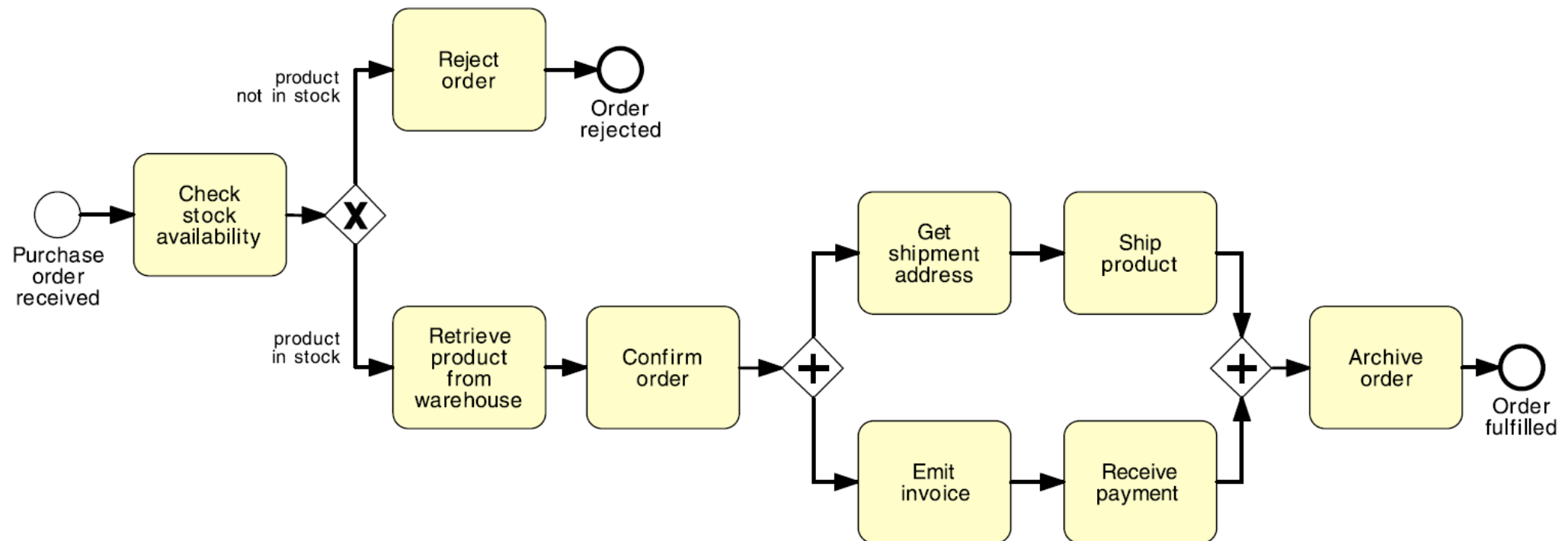


A Solution Without OR?

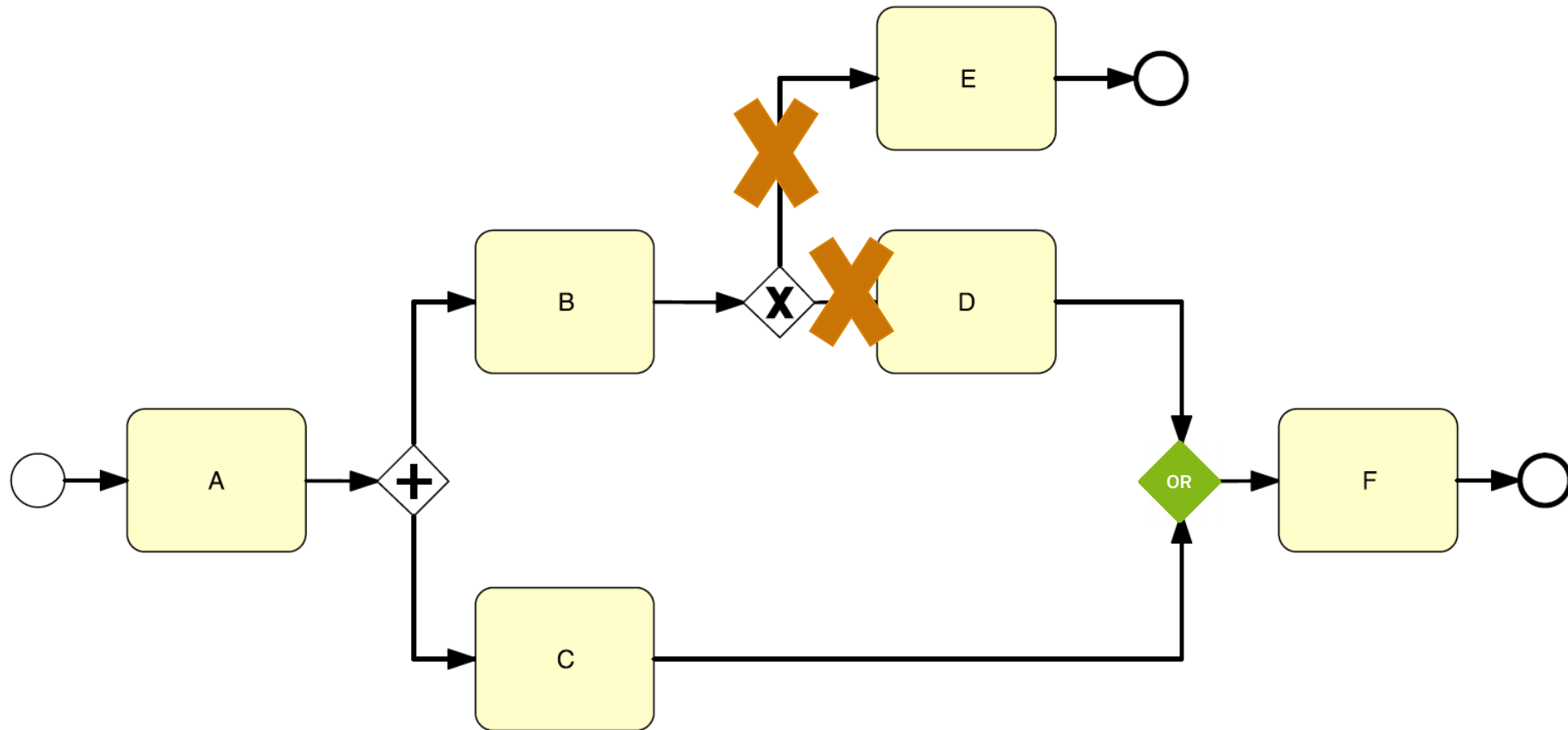
- **Yes!** This is it!
- But it is very cumbersome for 3+ branches!



Token Flow and Implicit Termination

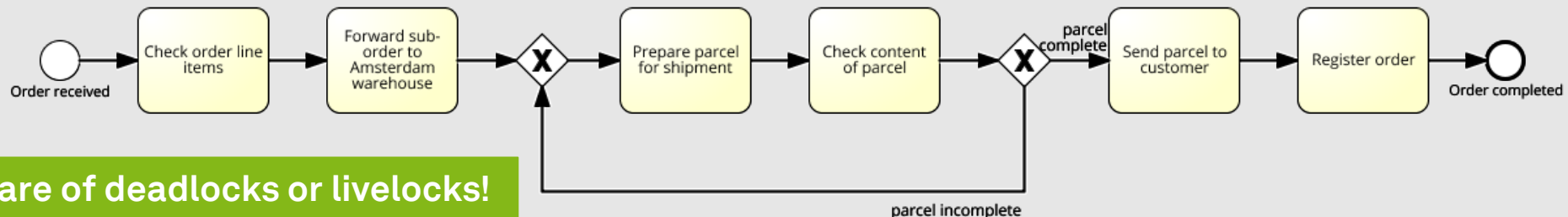
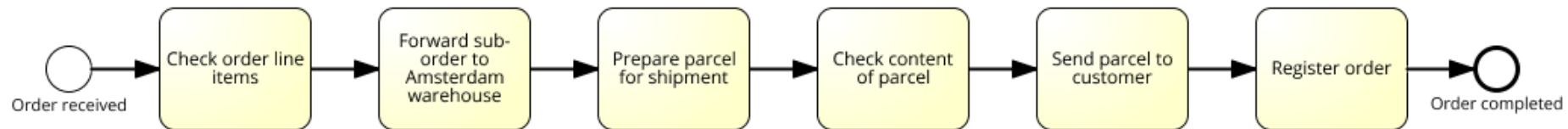


How Do We Fix This?



Rework Blocks

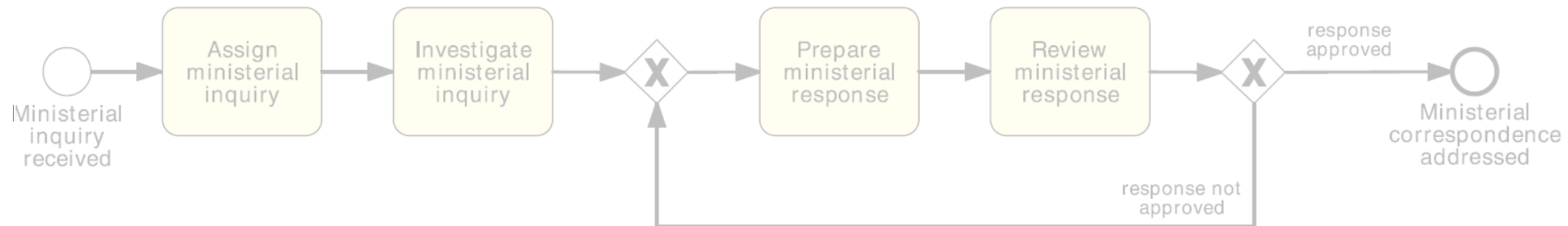
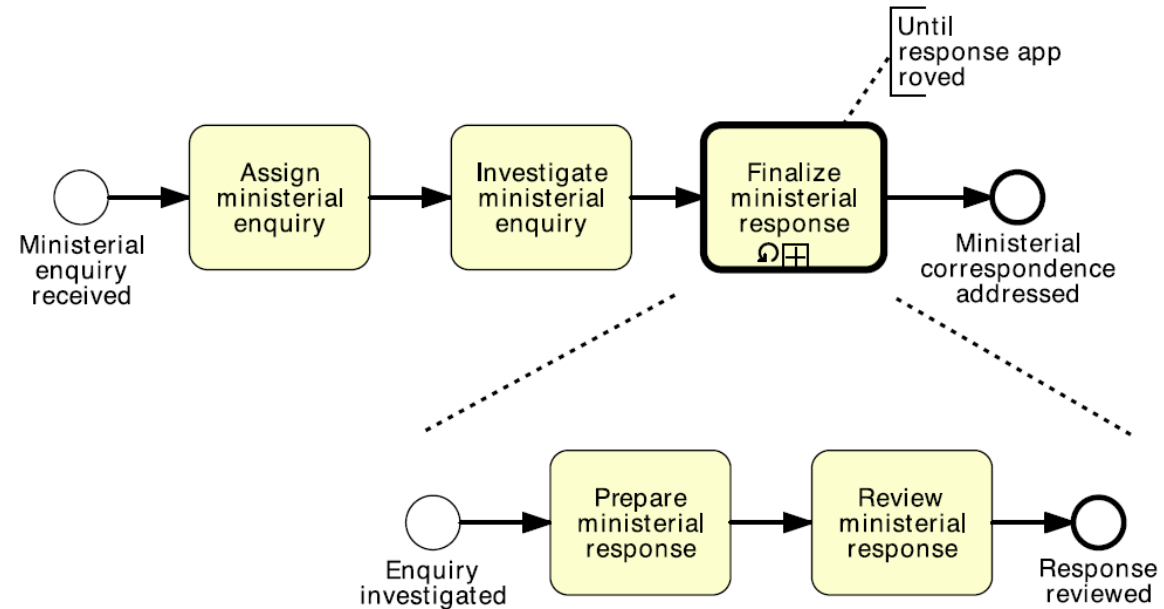
- Processes might need to **rework** one or several activities
- Imagine the following process model
- How do we **model** rework?
 - Activity “Check content of parcel” failed
 - Activity “Prepare parcel for shipment” needs to be reworked



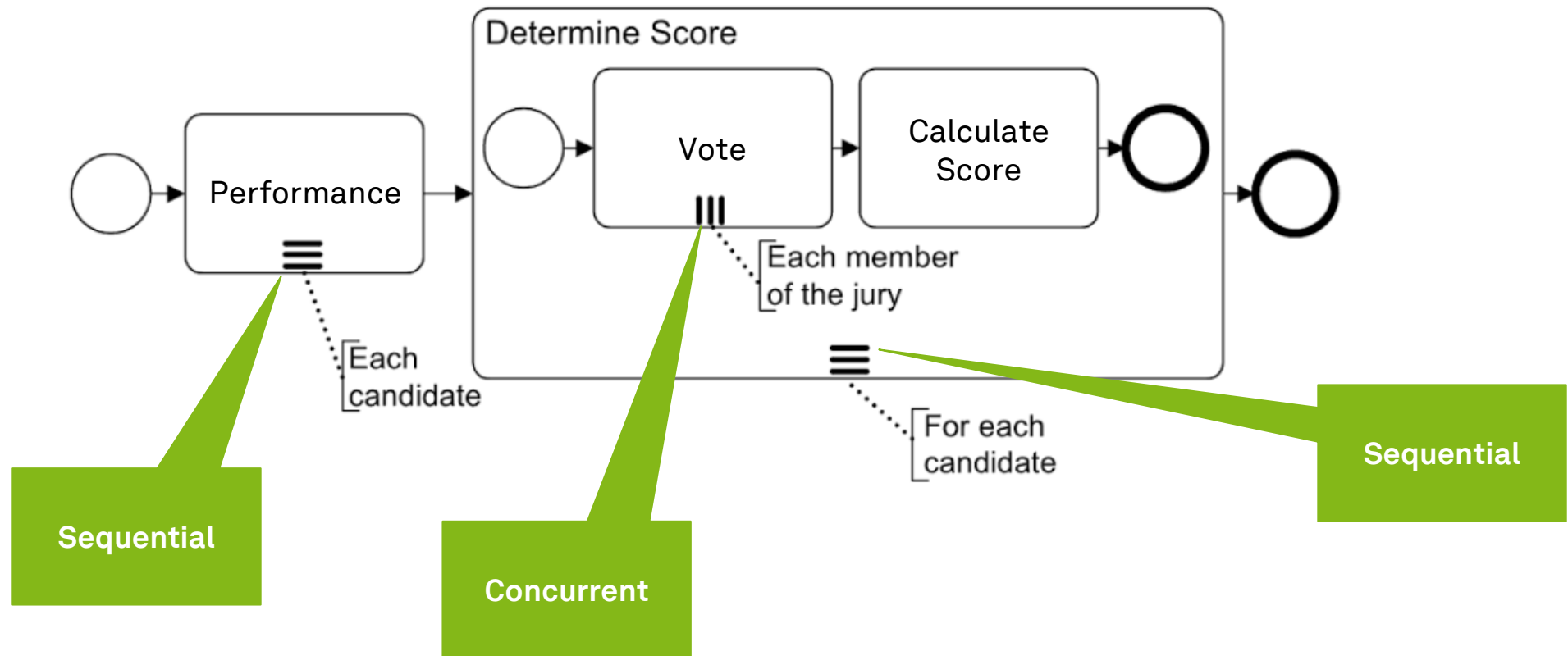
Be aware of deadlocks or livelocks!

Loop Activities

- For loops like these
 - use **XOR**,
 - not OR
 - **never AND!**

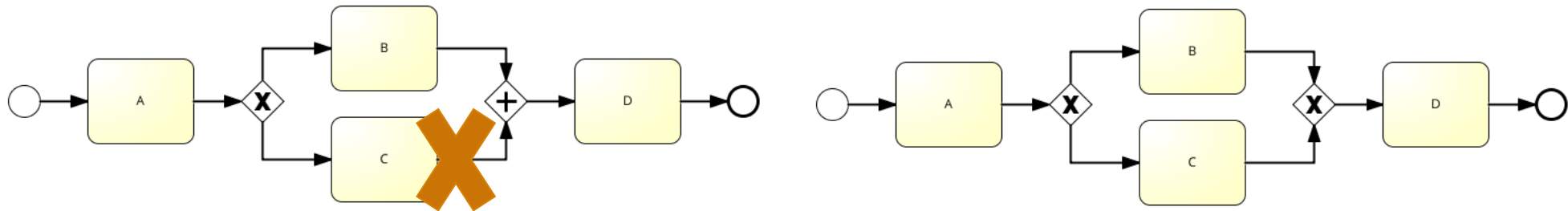


Multi-Instance Activities



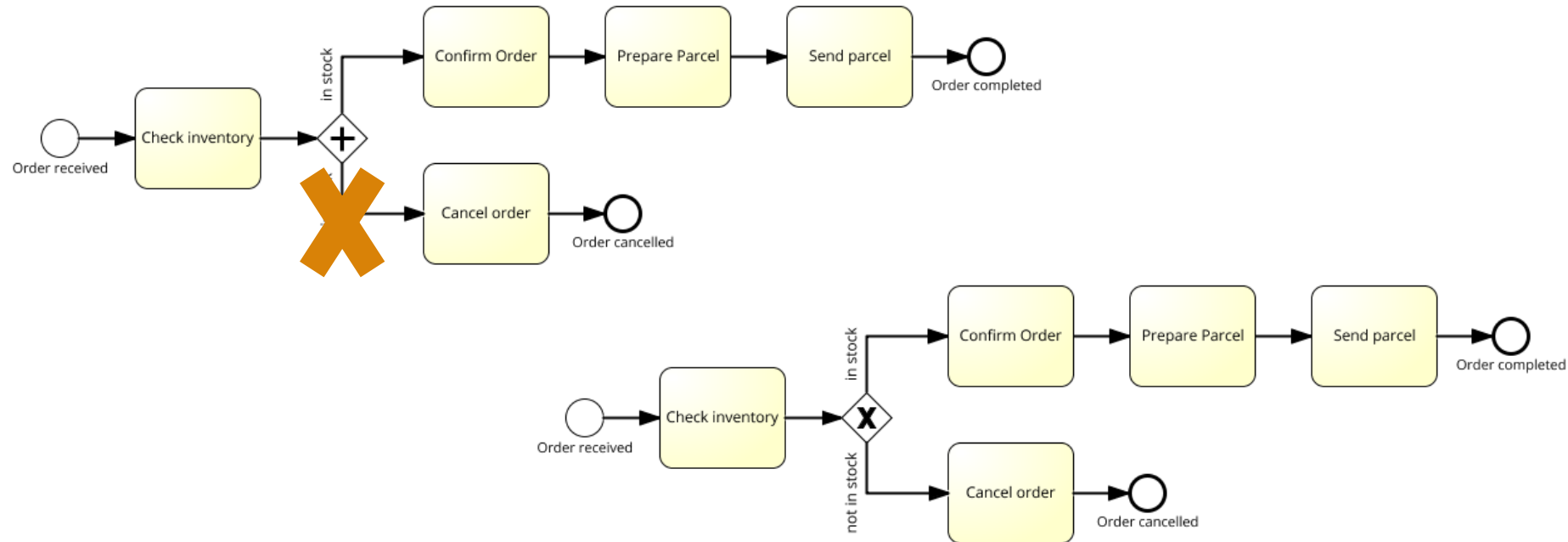
BPMN Use – Syntax

- Includes constructs and the **set of rules** to combine these constructs
- In linguistics, you determine the **structure of sentences** in a given language



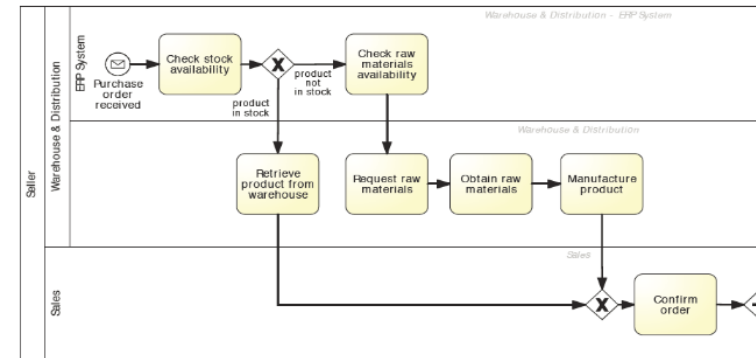
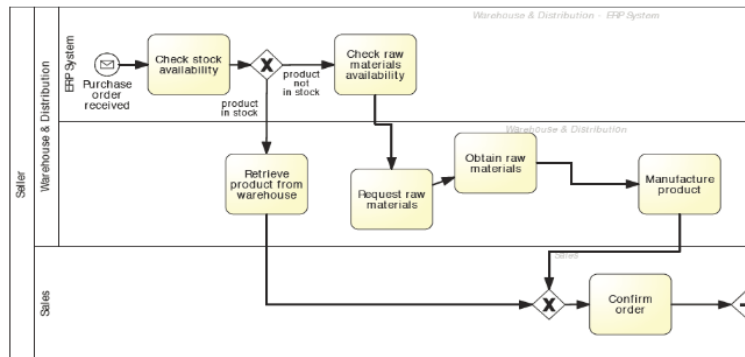
BPMN Use – Semantics

- Focusses on the study of **meaning**, and the study of **relations** between different elements.



BPMN Use – Pragmatics

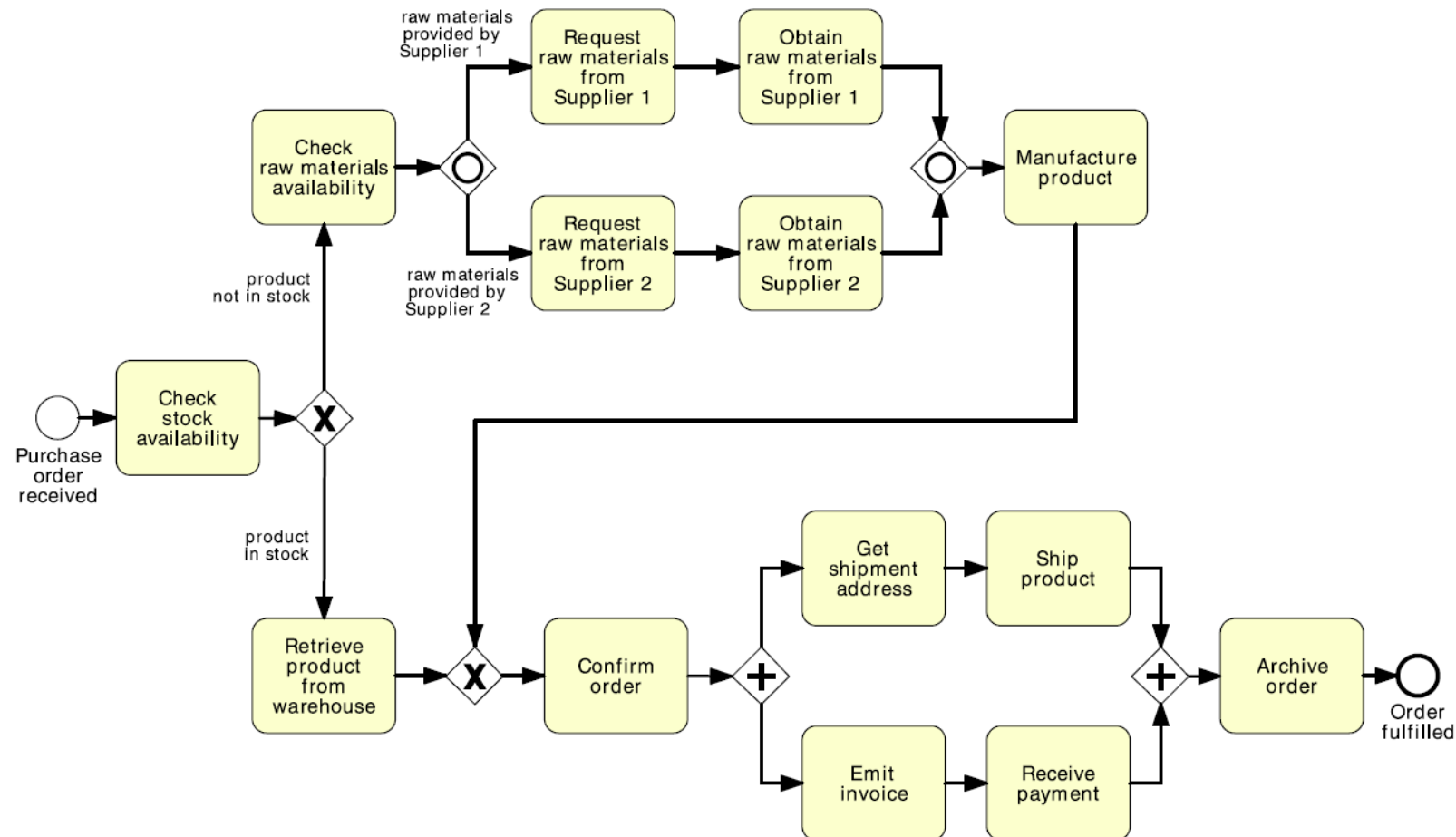
- Modeling using BPMN is from **top-left to bottom right**
- Do not cross arrows unless it is unavoidable



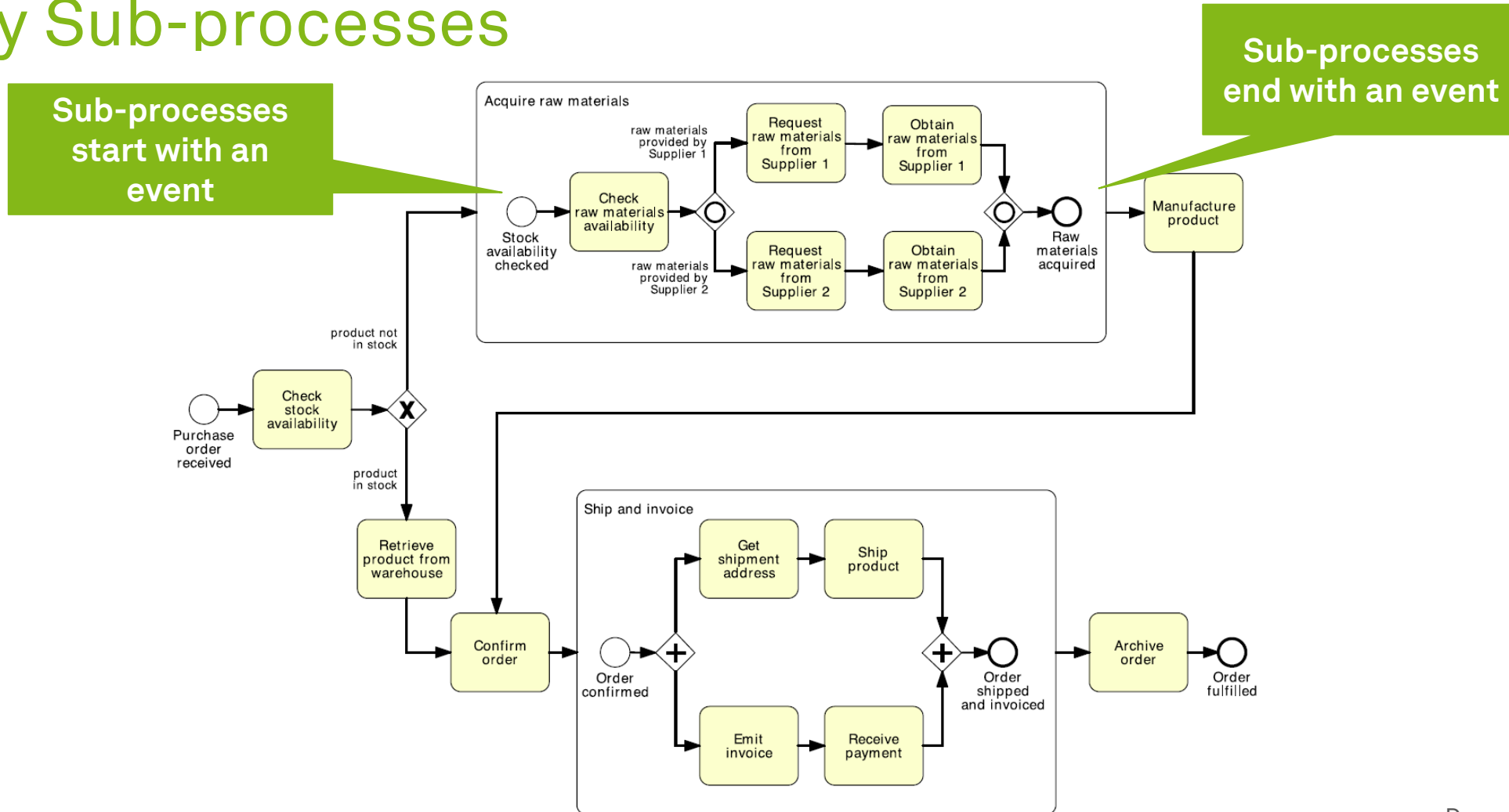
Decomposition, Resources, and Collaborations



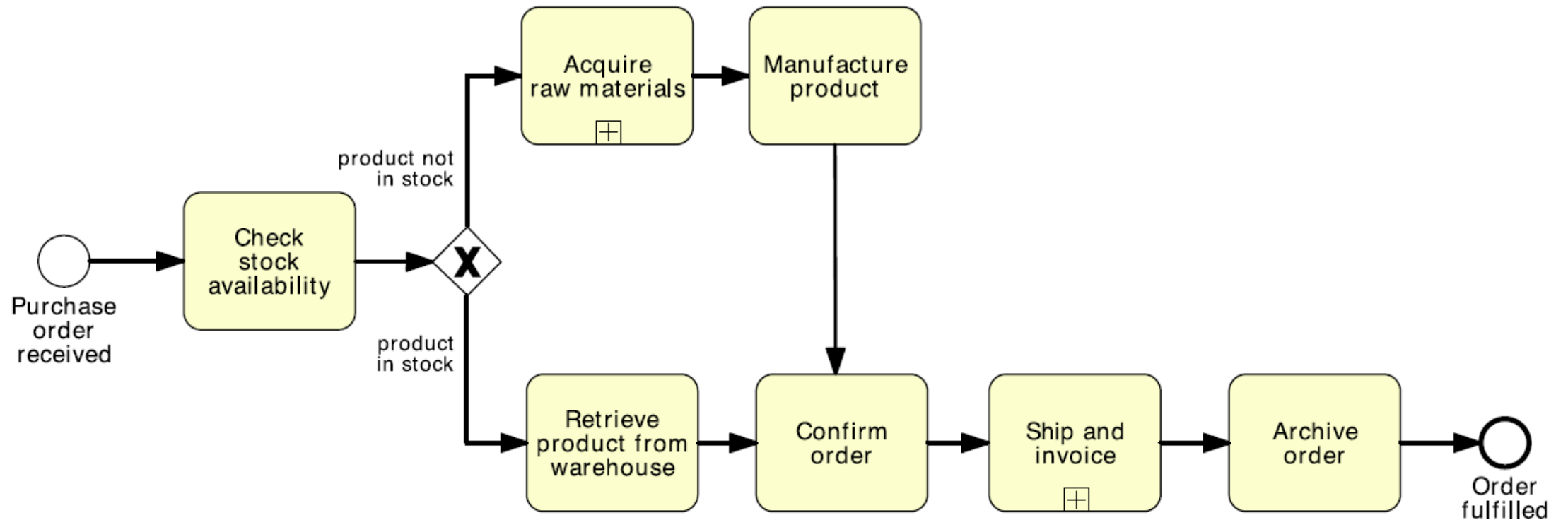
Process Decomposition



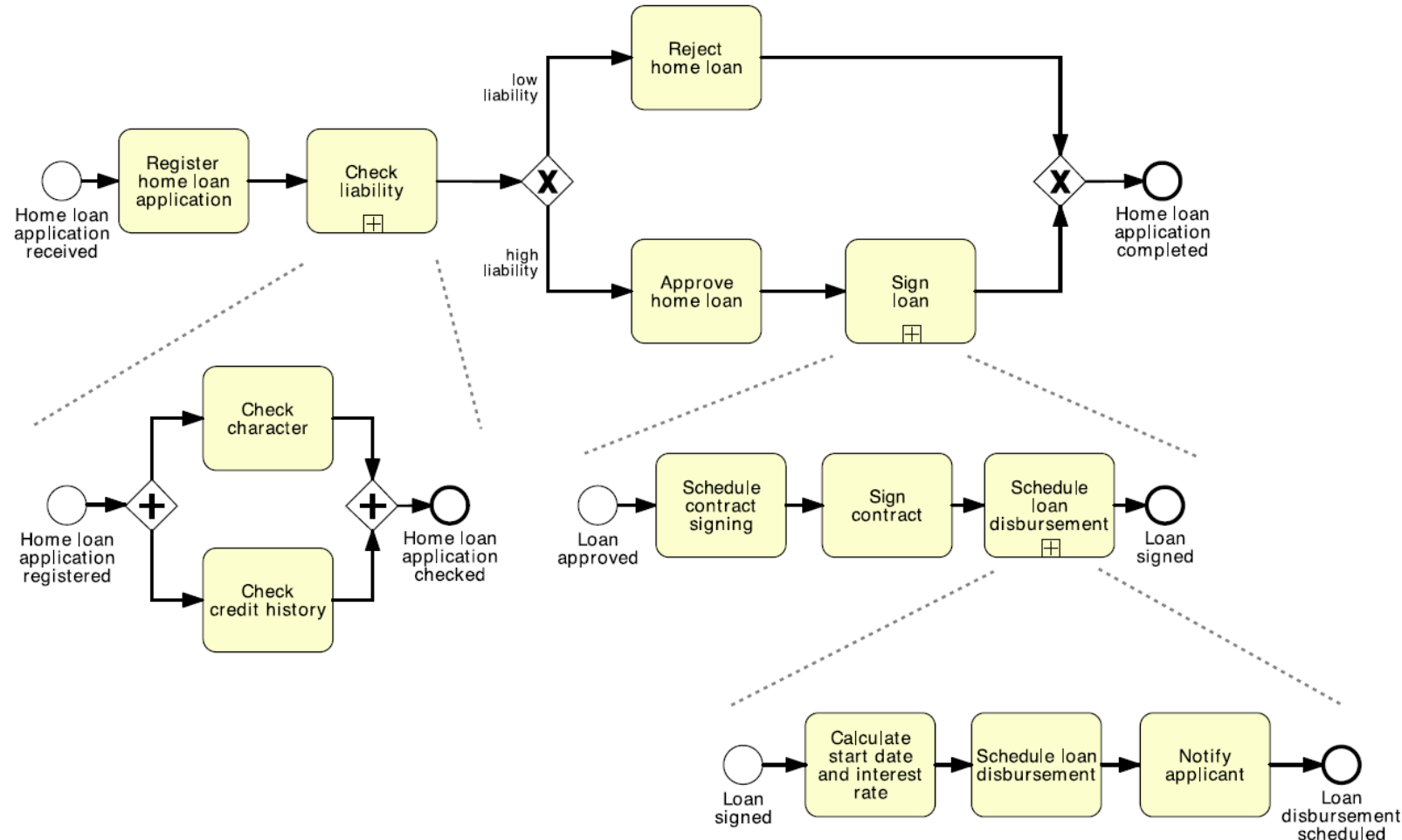
Identify Sub-processes



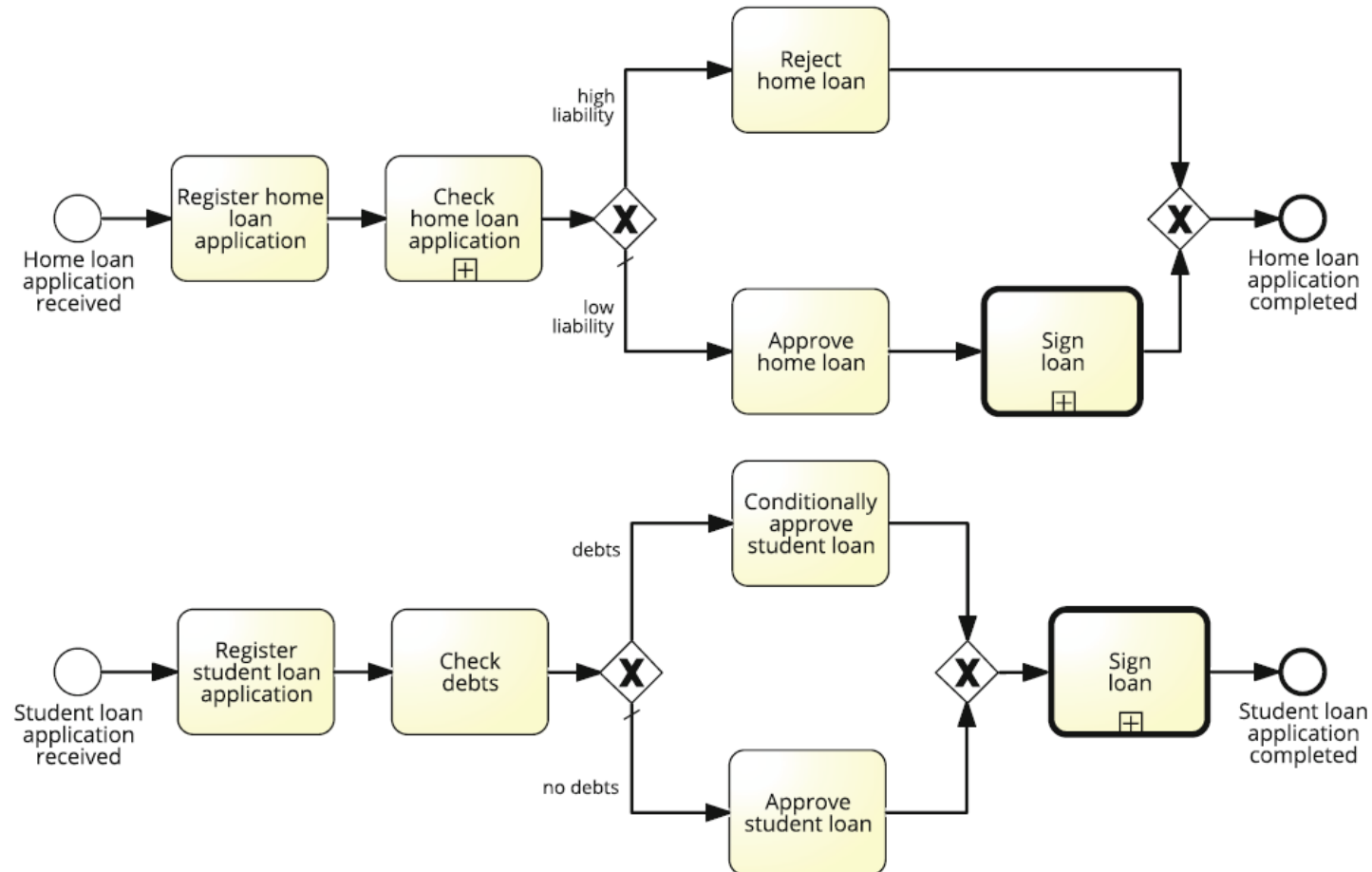
Process with Collapsed Activities



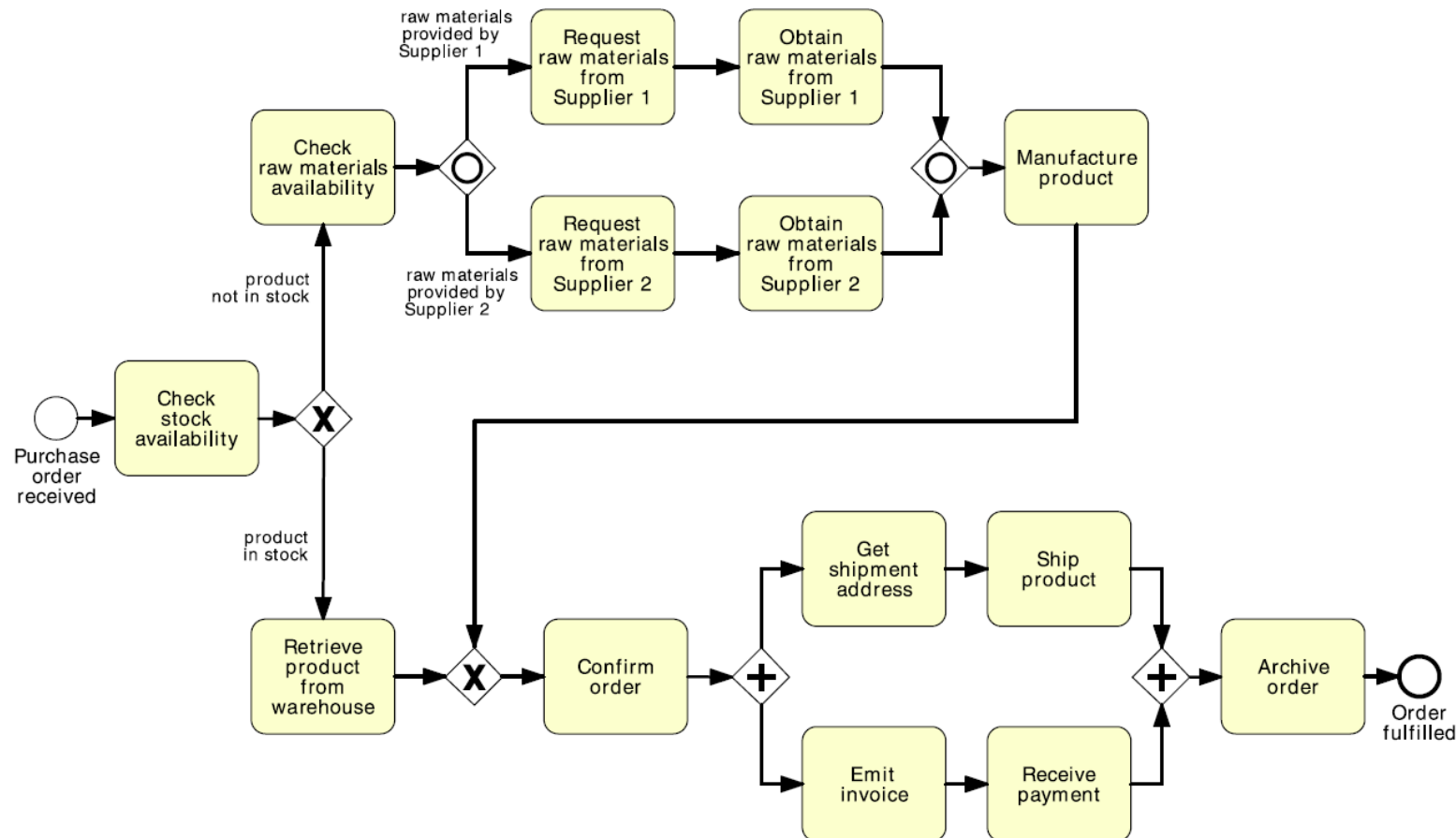
Hierarchically Decomposed Process



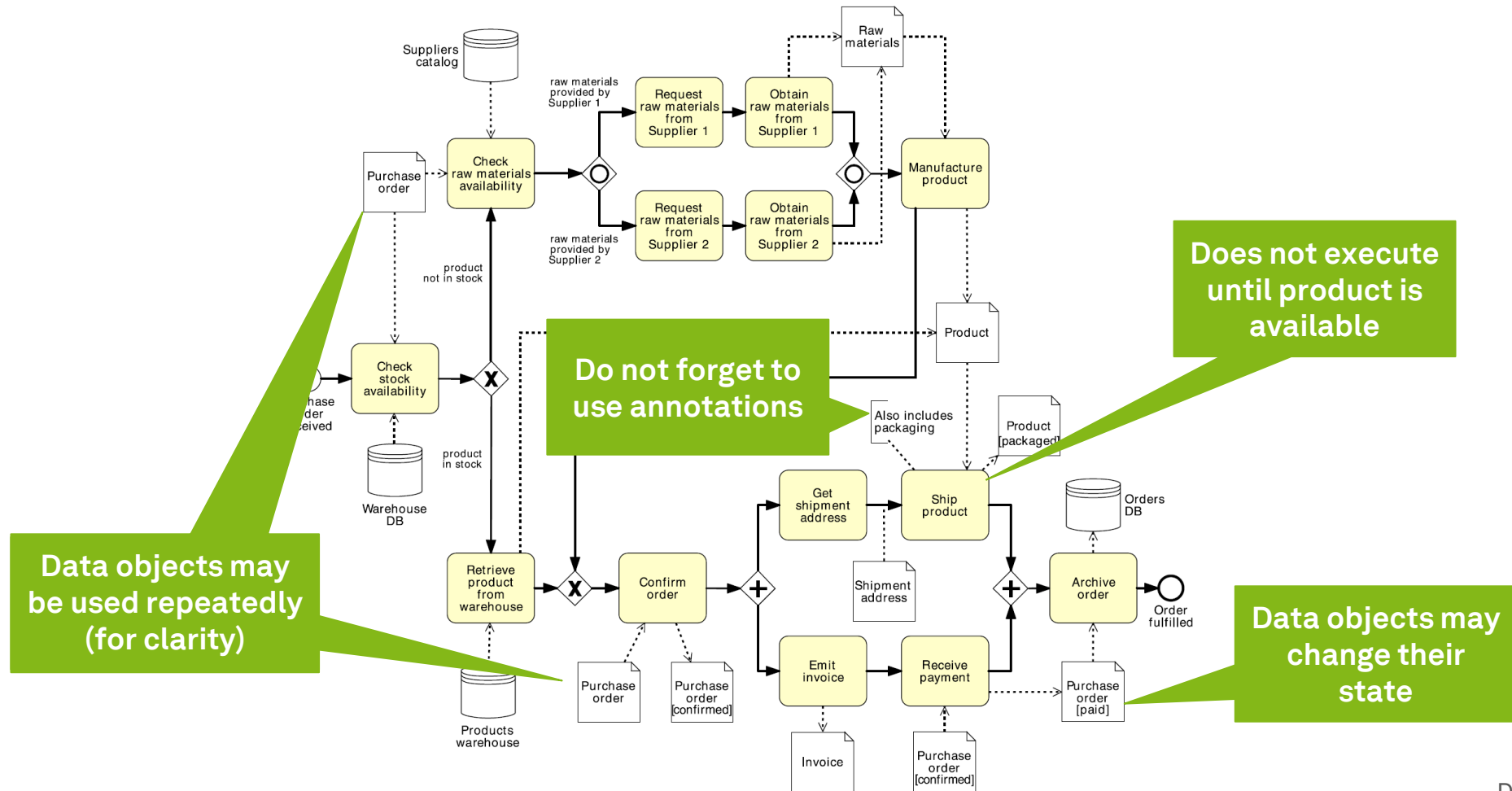
Call Activities



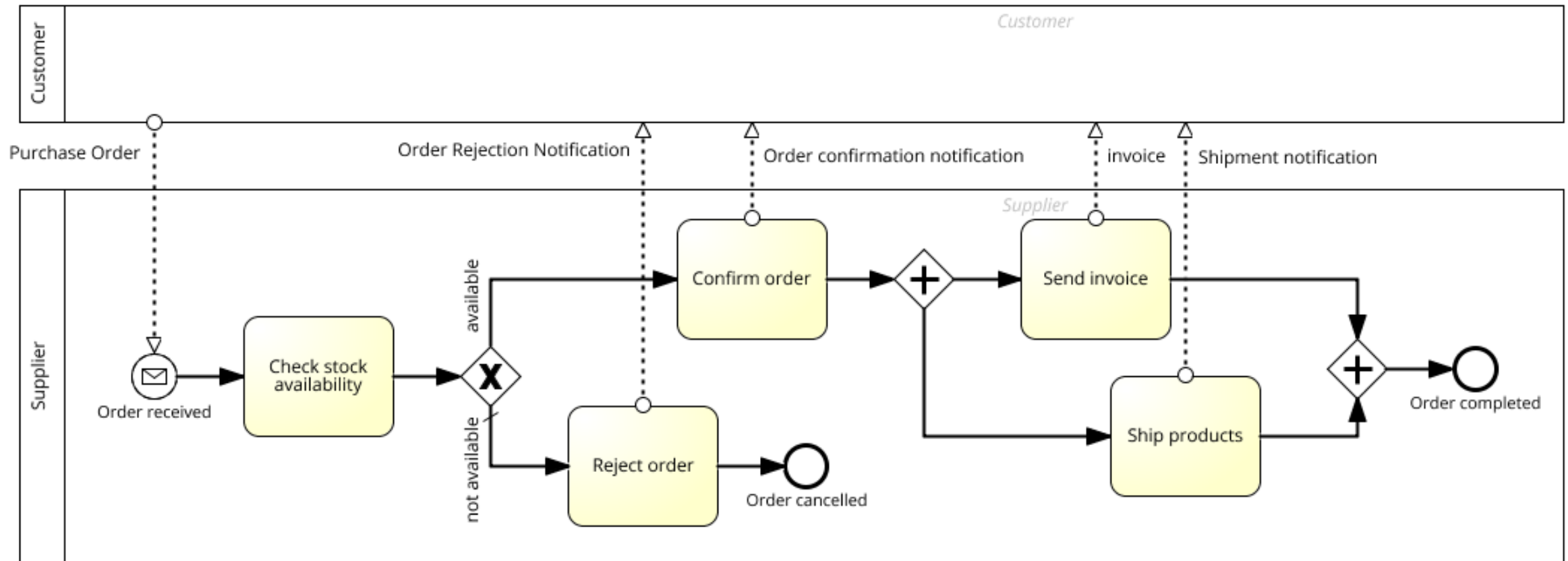
“But what about documents and stuff?”



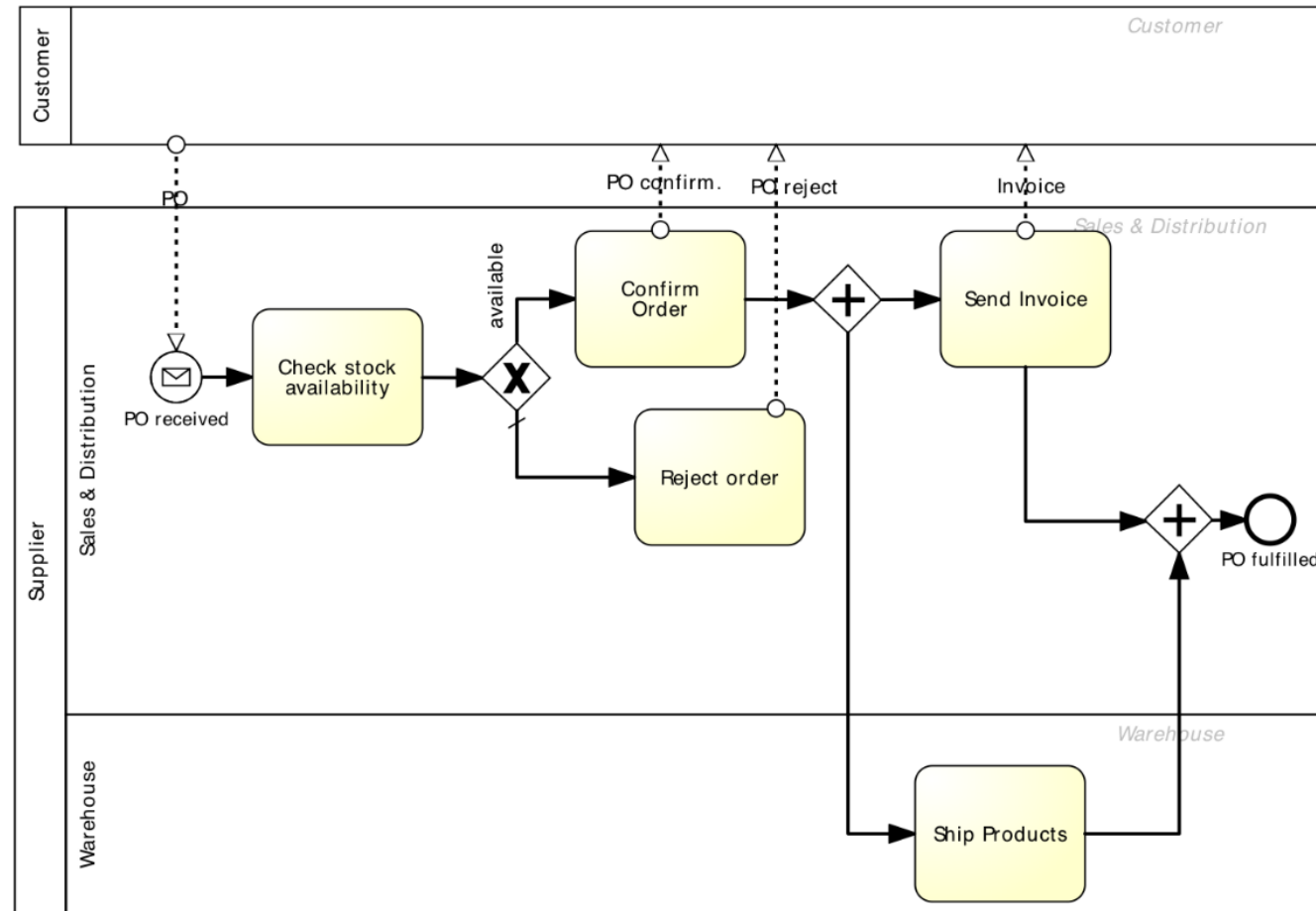
Passive Resources: The Data Perspective



Active Resources: The Organizational Perspective

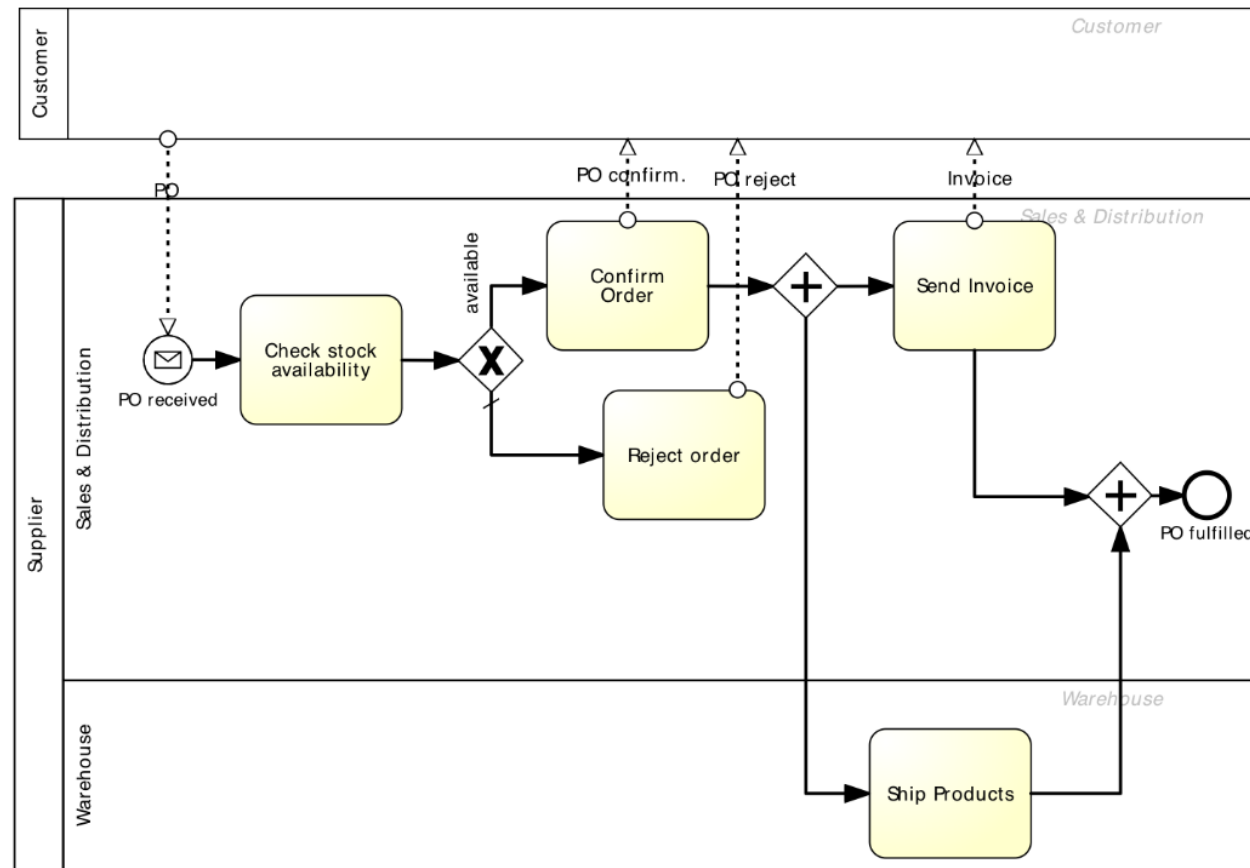


Order Management Process with Pools and Lanes



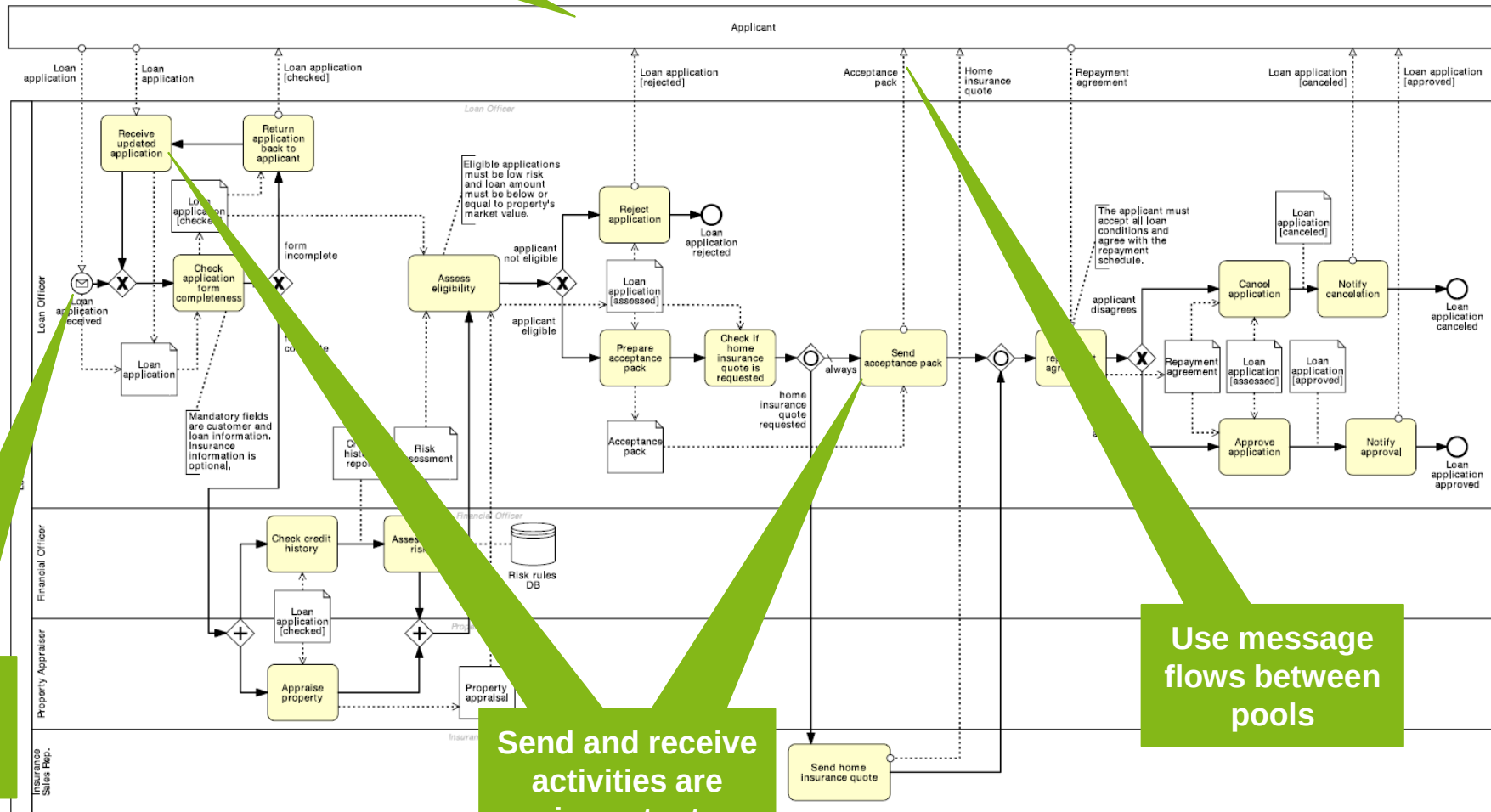
Modeling a Pool – Black Box vs. White Box

- When do we use a **black box** and when do we use a **white box**?



Collaborations

Private processes are collapsed pools



If the start is a message use a message event

Send and receive activities are important

Use message flows between pools

Learning Goals of Today

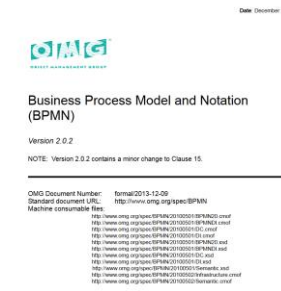
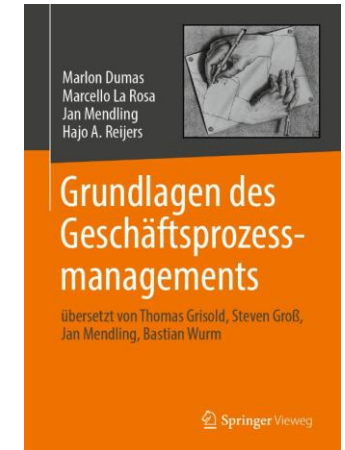
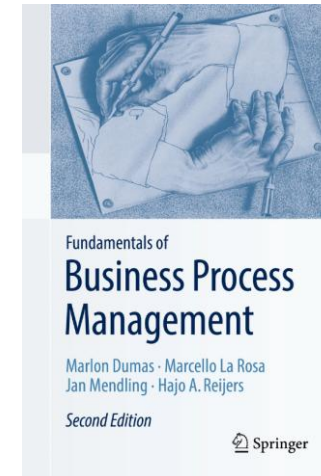
- Better understand the syntactics and semantics of BPMN
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Recap Questions

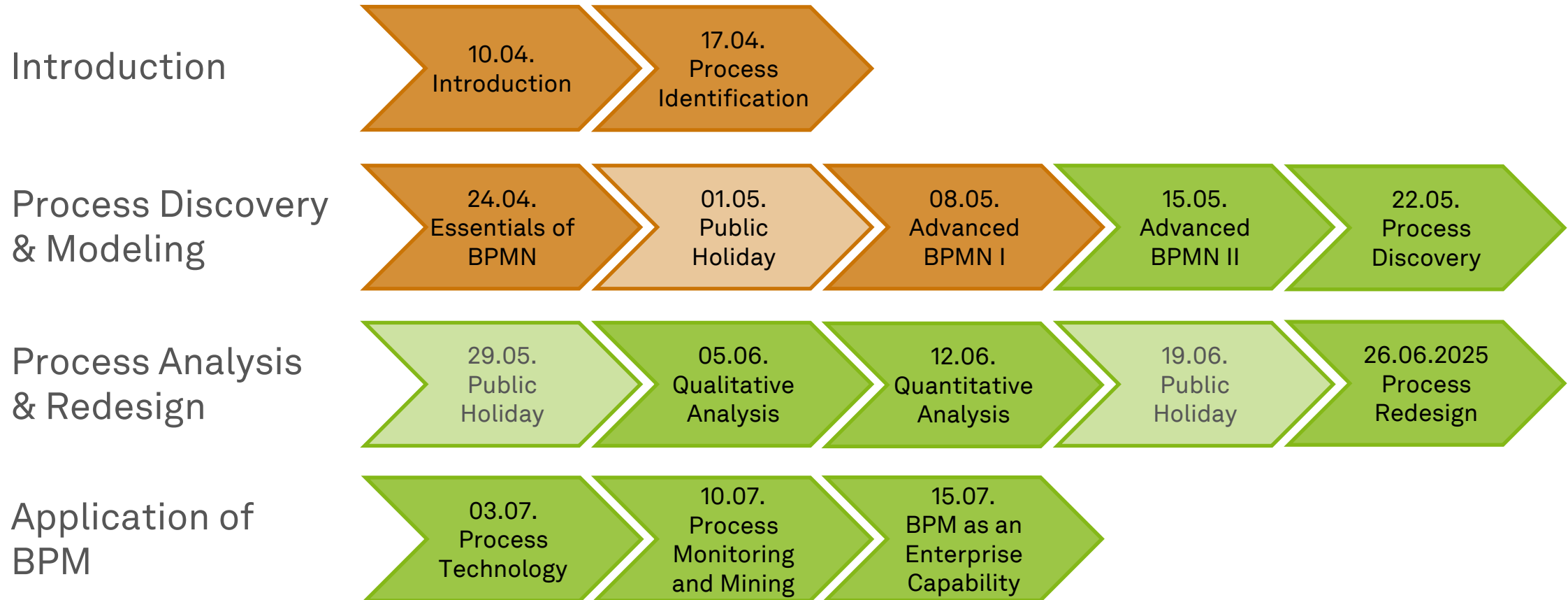
- Can we model everything without the inclusive-OR construct in BPMN?
- What is a loop, how do we model them in BPMN, and why can they cause deadlocks or even livelocks?
- What is process decomposition and its relation to private pools in BPMN?

Literature

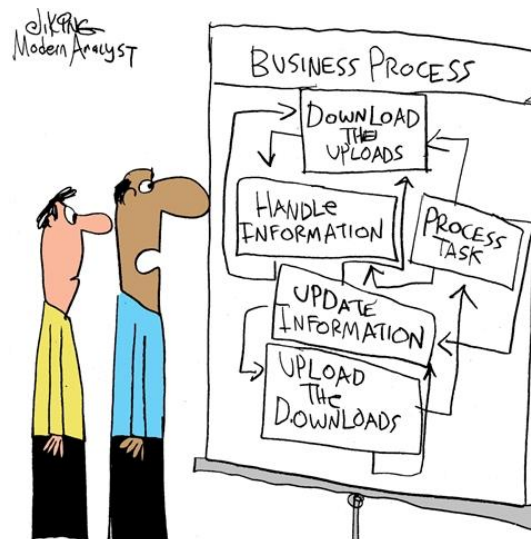
- **Section 3:** Essential Process Modeling
- **Section 4:** Advanced Process Modeling
- Business Process Model and Notation (BPMN)
 - <https://www.omg.org/spec/BPMN/2.0.2>



Roadmap



Any Questions about Process Modeling?



*"Charlie, did you really get these details
from the customer?"*



ec.cs.tu-dortmund.de

